HDOC DAY 54

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, arr[], and an integer k such that 0 < k < size. Pass the array to a function named max_subarray_sum() using its address. Use pointers to pass the array between the calling function and max_subarray_sum(). The function max_subarray_sum() must:

- Identify all contiguous subarrays of size k in arr[].
- Calculate the sum of the elements in each subarray.
- Find and display the maximum subarray sum.
- Display -1 if k is invalid.

CODE

```
include <stdio.h>

#include <stdlib.h>


void max_subarray_sum(int *arr, int size, int k)

{

  if (k <= 0 || k >= size)

  {

    printf("-1");

    return;

  }


  int sum = 0, max_sum;


  for (int i = 0; i < k; i++)

  {
```

```
        sum += *(arr + i);
    }

    max_sum = sum;

    for (int i = k; i < size; i++)
    {
        sum = sum + *(arr + i) - *(arr + i - k);

        if (sum > max_sum)
        {
            max_sum = sum;
        }
    }

    printf("%d", max_sum);
}

int main()
{
    int size, k;

    scanf("%d", &size);

    if (size <= 0)
        return 0;

    int *arr = (int *)malloc(size * sizeof(int));
```

```
    if (arr == NULL)
        return 1;

    for (int i = 0; i < size; i++)
    {
        scanf("%d", arr + i);
    }

    scanf("%d", &k);

    max_subarray_sum(arr, size, k);

    free(arr);

    return 0;
}
```